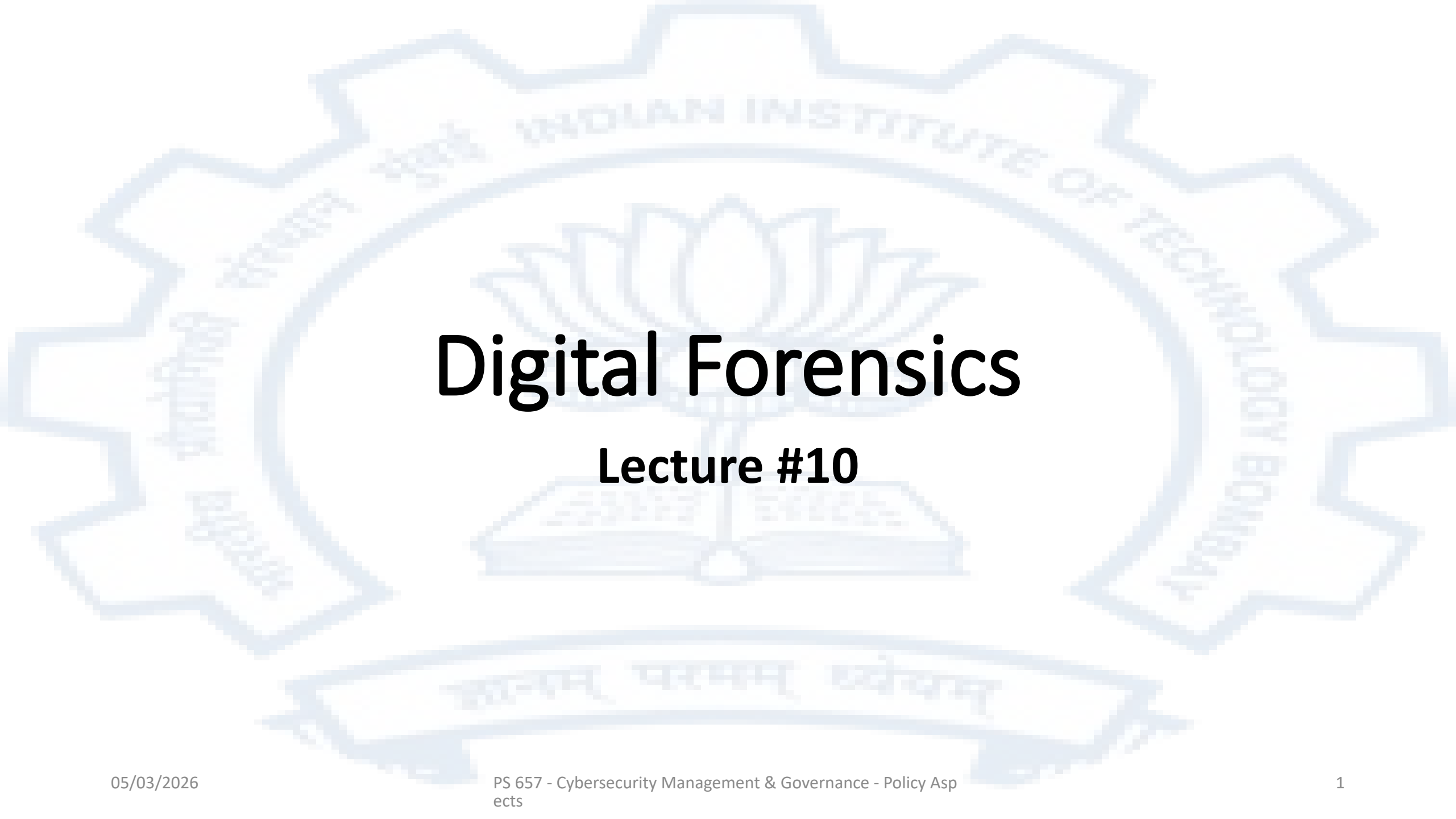




Digital Forensics

Lecture #10

Forensics

Locard's exchange principle: “*Every contact leaves a trace*”

Forensics refers to the use of science to (partially) reconstruct an incident from the traces that it leaves, and use that reconstruction to establish facts, relationships and sequences of events that may be offered as proof in a court of law.

Digital Traces

- In the domain of information processing, it is not inevitable that processing will leave digital traces and therefore directly applying the learning and experience in the general forensics domain may not always be appropriate. Though digital traces may often exist they are usually there as a result of conscious engineering decisions that may not have been taken to facilitate forensics
- A digital trace may be
 - Implicit
 - A low level examination of a disk may reveal a certain type of content that does not belong to any file that is currently in the file system. This may lead to a conclusion that there was a file that has now been deleted that did contain this content
 - Explicit
 - Many processes create timestamped entries (logs) when certain events occur such as a failed login or opening a network connection etc.

Digital Forensics – The DFRWS definition

In 2001 the Digital Forensics Research Workshop proposed the following definition

Digital forensics *is the use of scientifically derived and proven methods toward the preservation, collection, validation, identification, analysis, interpretation, documentation and presentation of digital evidence derived from digital sources for the purpose of facilitating or furthering the reconstruction of events found to be criminal, or helping to anticipate unauthorized actions shown to be disruptive to planned operations.*

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Digital Forensics – The NIST definition

A few years after the DFRWS definition, NIST proposed the following definition

Digital forensics (NIST) is considered the application of science to the identification, collection, examination, and analysis of data while preserving the integrity of the information and maintaining a strict chain of custody for the data. Data refers to distinct pieces of digital information that have been formatted in a specific way.

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The Daugbert principle for forensic evidence

In the USA the Daugbert principle is used to determine whether forensic evidence of a particular kind is admissible in a particular case. US law gives judges considerable leeway in deciding on the admissibility of forensic evidence

1. The theoretical underpinnings of the methods must yield testable predictions by means of which the theory could be falsified.
2. The methods should preferably be published in a peer-reviewed journal.
3. There should be a known rate of error that can be used in evaluating the results.
4. The methods should be generally accepted within the relevant scientific community.

Admissibility of forensic evidence in Indian Courts

The issue of admissibility of forensic evidence in Indian courts is governed by Section 39 to 45 of Bharatiya Sakshya Adhiniyam, 2023

Opinion of Experts -Section 39(1)

Opinion of Examiner of Electronic Evidence- Section 39(2)

Opinion as to handwriting-Section 41(1)

Opinion as to electronic signature -Section 41(2)

Opinion as to existence of right or custom-Section 42

Opinion as to usages, tenets, etc -Section 43

Opinion on relationship -Section 44

Examiner of Electronic Evidence

CHAPTER XIIA of the Information Technology Act, 2000 empowers the Central Government under section 79A to notify any Department, body or agency of the Central Government or a State Government as an Examiner of Electronic Evidence for the purposes of providing expert opinion on electronic form evidence before any court or other authority specified by notification in the Official Gazette. The *Explanation* clause of section 79A further articulates that the “Electronic Form Evidence” means any information of probative value that is either stored or transmitted in electronic form and includes evidence, digital data, digital video, cell phones, digital fax machine etc.

Some organizations that have been accredited as Examiner of Electronic Evidence

1. Digital Forensic Laboratory, Cyber Defence Centre, National Forensic Sciences University (NFSU), Gujarat
2. Regional Forensic Science Laboratory (RFSL), Surat, Gujarat
3. Cyber Forensics & Digital Evidence Examiners Laboratory (CF&DEEL), Kolkata, West Bengal
4. Forensic Wing Lab, Defence Cyber Agency (DCyA), Rajaji Marg, New Delhi
5. Cyber Forensics Laboratory, Navy Cyber Group
6. Cyber Forensic Division, State Forensic Sciences Laboratory, Thiruvananthapuram, Kerala
7. Cyber Forensic Laboratory, Air Force Cyber Group, New Delhi
8. Cyber Forensic Lab, CERT-In
9. Regional Forensic Science Laboratory, Northern Range, Dharamshala, Himachal Pradesh
10. Cyber Forensic Laboratory, Army Cyber Group, DGM0, Signals Enclave, New Delhi
11. State Forensic Science Laboratory, Madiwala, Bangalore
12. Central Forensic Science Laboratory (CFSL), Hyderabad
13. Directorate of Forensic Science, Gandhi Nagar, Gujarat
14. Computer Forensic and Data Mining Laboratory (CFDML), Serious Fraud Investigation Office (SFIO), Delhi
15. Forensic Science Laboratory, Government of NCT, Rohini, New Delhi
16. Cyber Forensic Lab, Chennai

Digital Forensics – A current operational definition

The DFRWS definition while primarily focusing on criminal investigation also envisaged an anticipatory element which is relevant in scenarios of cyber defense. The NIST definition on the other hand is more law-centric.

In current times a more operational definition is proposed

Digital forensics *is the process of identifying and reconstructing the relevant sequence of events that have led to the currently observable state of a target IT system or (digital) artifacts.*

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Modelling the Digital Forensics Process

- Cognitive scientists do not seem to have studied the process of digital forensics in any detail or in any systematic way
- However there has been considerable study of the intelligence analysis process. In particular Pirolli and Card developed a process to “make sense” of intelligence

P. Pirolli and S. Card, “Sensemaking processes of intelligence analysts and possible leverage points as identified through cognitive task analysis,” in *Proceedings of the 2005 International Conference on Intelligence Analysis*, 2005.

- This model is claimed to be useful as a model to understand digital forensics in the absence of more specific work

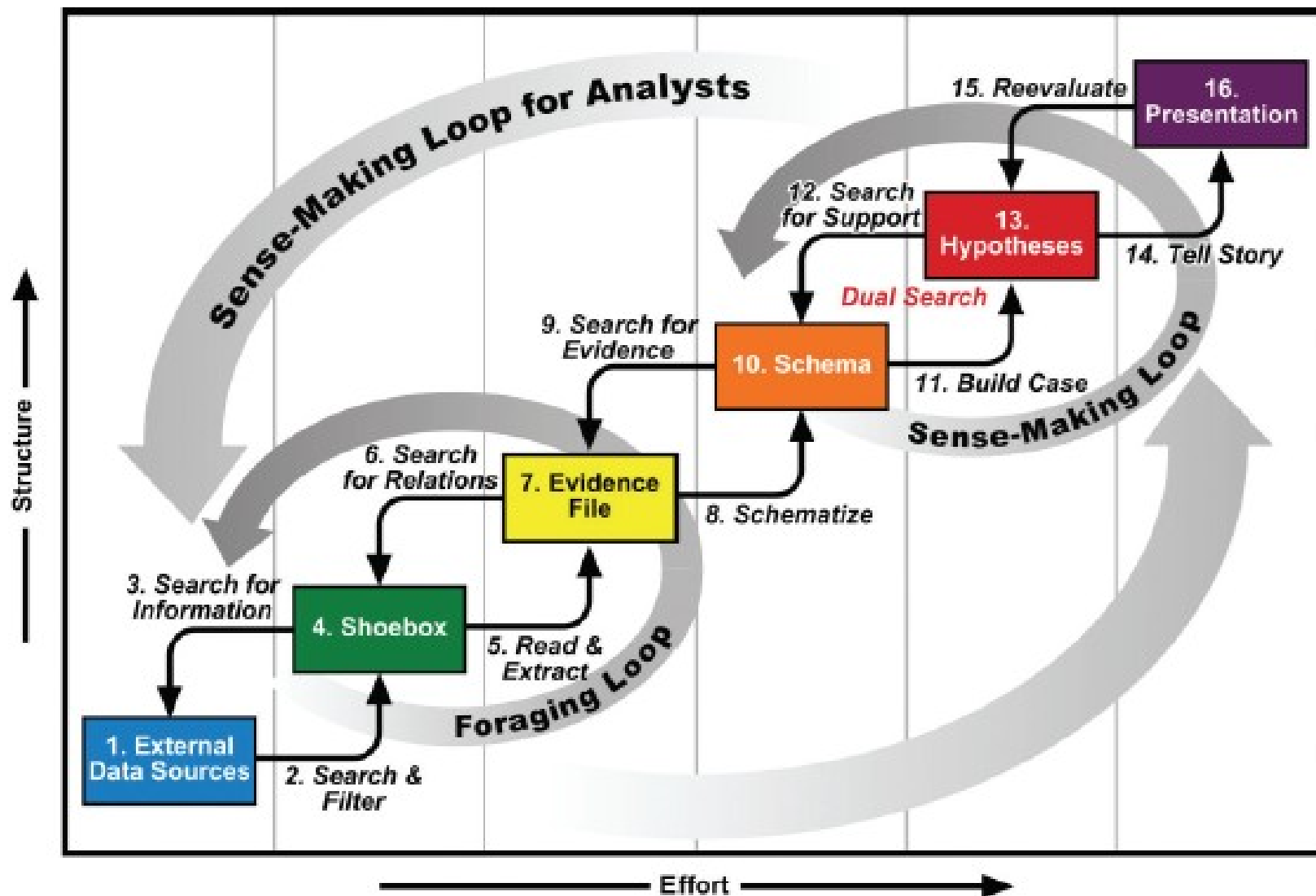


Figure 9.1: Notional model of sense-making loop for analysts derived from cognitive task analysis [933, p.44].

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The Foraging Loop

- ***External data sources*** include all potential evidence sources for a specific investigation such as disk images, memory snapshots, network captures and reference databases such as hashes of known files.
- The ***shoebox*** is a subset of all the data that have been identified as potentially relevant, such as all the email communications between two persons of interest. At any given time, the contents of the shoebox can be viewed as the analyst's approximation of the information content that is potentially relevant to the case.
- The ***evidence file*** contains only the parts that are directly relevant to the case such as specific email exchanges on a topic of interest.

The Sense-Making Loop

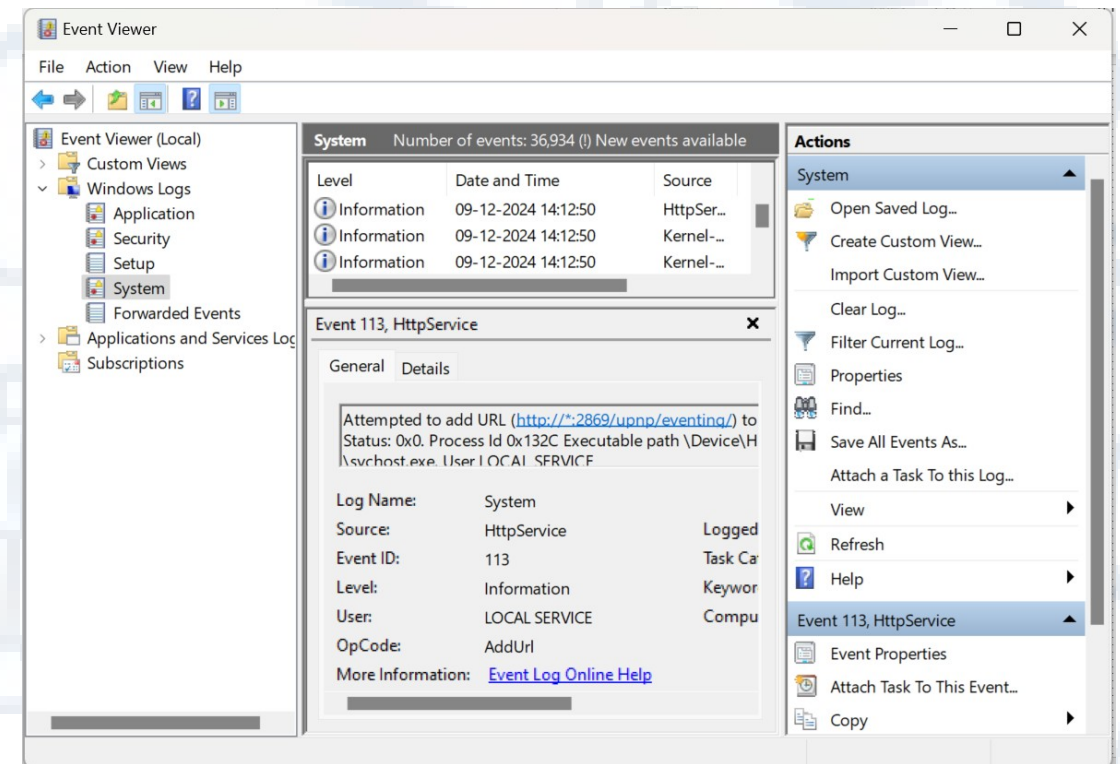
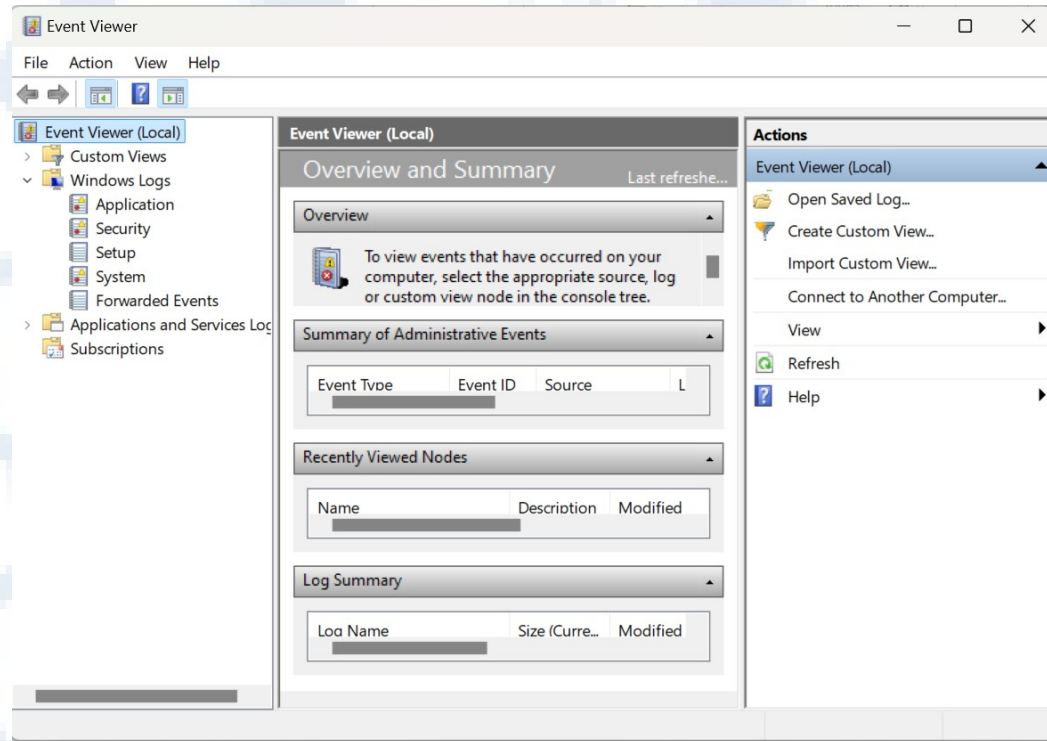
- The *schema* contains a more organised version of the evidence such as a timeline of events or a graph of relationships, which allows higher-level reasoning over the evidence.
- A *hypothesis* is a tentative conclusion that explains the observed evidence in the schema and, by extension, could form the final conclusion.
- Once the analyst is satisfied that the hypothesis is supported by the evidence, the hypothesis turns into a *presentation*, which is the final product of the process. The presentation usually takes on the form of an investigator's report that both speaks to the high-level conclusions that are relevant to the legal case and also documents the low-level technical steps based on which the conclusion has been formed.

Digital Traces

Digital traces are not an inevitable consequence of digital processing. Various teams and individuals make uncoordinated engineering decisions that result in digital traces and these traces may or may not be sufficient to (partially) reconstruct incidents.

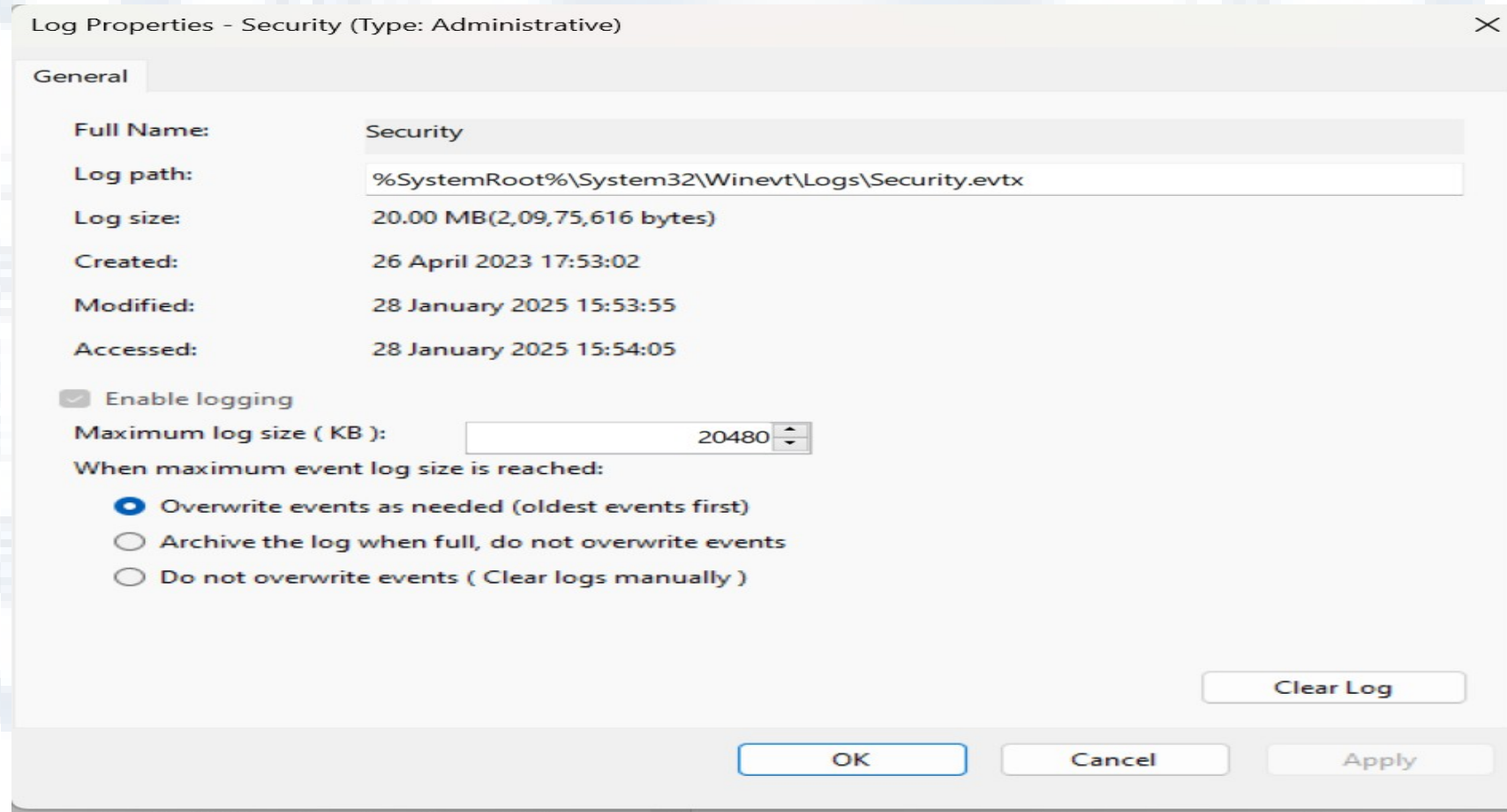
This is in contrast with conventional forensics where the laws of nature are the “operating system” and every action is subject to the laws of nature at every point of time and space.

Windows logging system



The implementers of every application need to make decisions on what to log and in what detail
The operating system has some pre-defined granularity which many not be changeable

Log management



Even though logging may be enabled at the right degree of granularity, improper log management Practices may result in the needed entries being unavailable

Operating System Analysis

The major aspects of Operating Systems analysis from the forensics viewpoint are

1. Storage forensics
2. Data acquisition
3. Filesystem analysis
4. Block device analysis
5. Data recovery and file content carving

Storage Forensics

- Persistent storage is the primary source of evidence for most digital forensics investigations
- Persistent storage is usually
 - HDD (Hard disk drives)
 - SSDs (Solid state disks)
 - Optical drives (now relatively rare)
 - Tape drives (now relatively rare)
- The operating system sees data as a hierarchy of abstractions
 - Physical media
 - Block device
 - File system
 - Application layer abstractions

Data Acquisition

Data acquisition may take place at the physical or logical level

- Physical data acquisition
 - This is the process of obtaining data directly from the hardware media without the use of any (untrusted) software. An example is to desolder memory chips from a mobile phone and read data directly from the chips. This gives a high degree of trust but may be very expensive and may be destructive
 - Sometimes a Host Bus Adapter protocol may be used to interact with the device and recover a copy. This type of acquisition is sometimes called a pseudo-physical acquisition
- Logical data acquisition
 - This is the process of using one or more software layers or protocols to acquire data. The faithfulness of this technique depends upon the correctness of the tools used.

Block Device Analysis

- The operating system usually sees raw devices as a collection of blocks.
- The operating system can read or write a whole block at a time. Traditionally for hard disks a block was usually 512 bytes and now is often 4096 bytes
- Each device can be considered to have a number of partitions where each partition is a contiguous number of blocks.
- A number of partitions can be organized into logical volumes

File System Analysis

- A **file** is a named (opaque) sequence of bytes that is stored persistently. The interpretation of the format and content of a file is almost always out of the purview of the operating system and is a matter for applications that use the file.
- A **filesystem** is an OS subsystem that is responsible for the persistent storage and organization of user and system files.
- Filesystem forensics uses knowledge of the filesystem datastructures and algorithms to
 - Retrieve data in files independent of the filesystem instance where they were created
 - Retrieve fragments of data which may not be accessible via the standard filesystem APIs

Data Recovery & File Content Carving

- One of the earliest data recovery tools was the file undelete functionality
- The undelete functionality was possible because when a file is deleted, its blocks are marked as deleted but not overwritten. Eventually these marked blocks will get allocated to some other file and then get overwritten, but till then remain available
- File content carving is the process of recovering and reconstructing file content from block storage without using the filesystem metadata. More generally, data (structure) carving is the process of reconstructing logical objects (such as files and database records) from a bulk data capture (disk/RAM image) without using metadata that describes the location and layout of the artifacts.

File Carving Illustration

Based upon two simple observations

- a). Most file formats have a specific beginning and end of file marker
- b). Most filesystems prefer sequential allocation of blocks to files

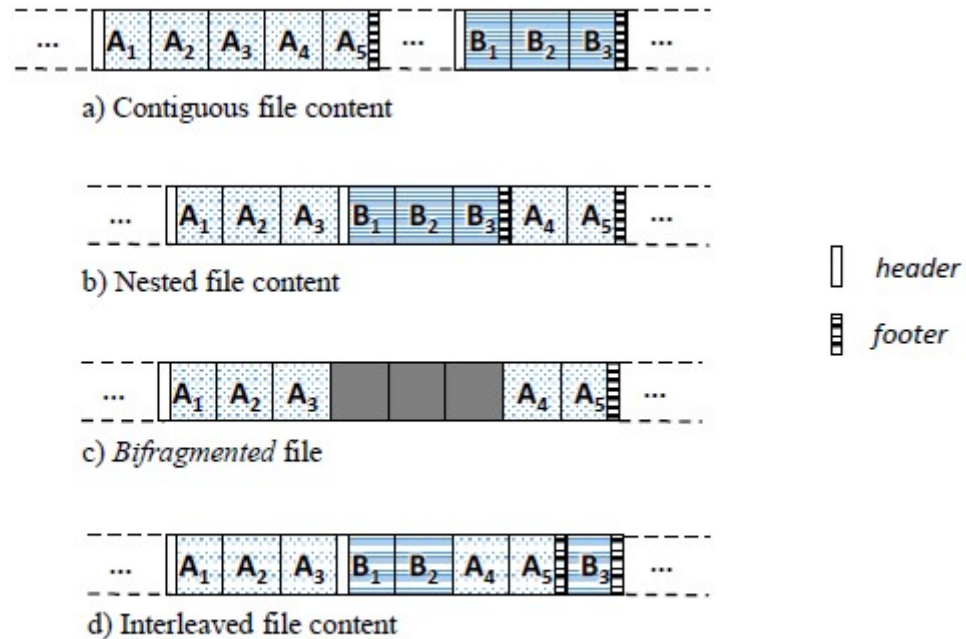


Figure 9.2: Common file content layout encountered during carving.

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Main Memory Forensics

The early view of best forensic practices was to literally pull the plug on a machine that was to be impounded. The rationale was that this would remove any possibility of alerting the processes running on the host and would preempt any attempts to hide information. Over time, experience has shown that these concerns were largely exaggerated and that the substantial and irreversible loss of important forensic information such as open connections and encryption keys was rarely justified. Studies have clearly demonstrated that data tend to persist for a long time in volatile memory ([950], [951]). There is a wealth of information about a system's run-time state that can be readily extracted, even from a snapshot [949]

Main memory analysis can give information about

- Running processes
- Open files
- Open network connections
- Artifacts and fragments

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The issue is how to capture a main memory image

- Live system
- Snapshot

Application Forensics

The great advantage here is that a direct observation of user interactions is possible. For example in a web browser there are the following interesting sources of data for forensic analysis

- URL/Search history
- Form data
- Temporary files
- Downloaded files
- HTML 5 local storage
- Cookies

Cloud Computing Models

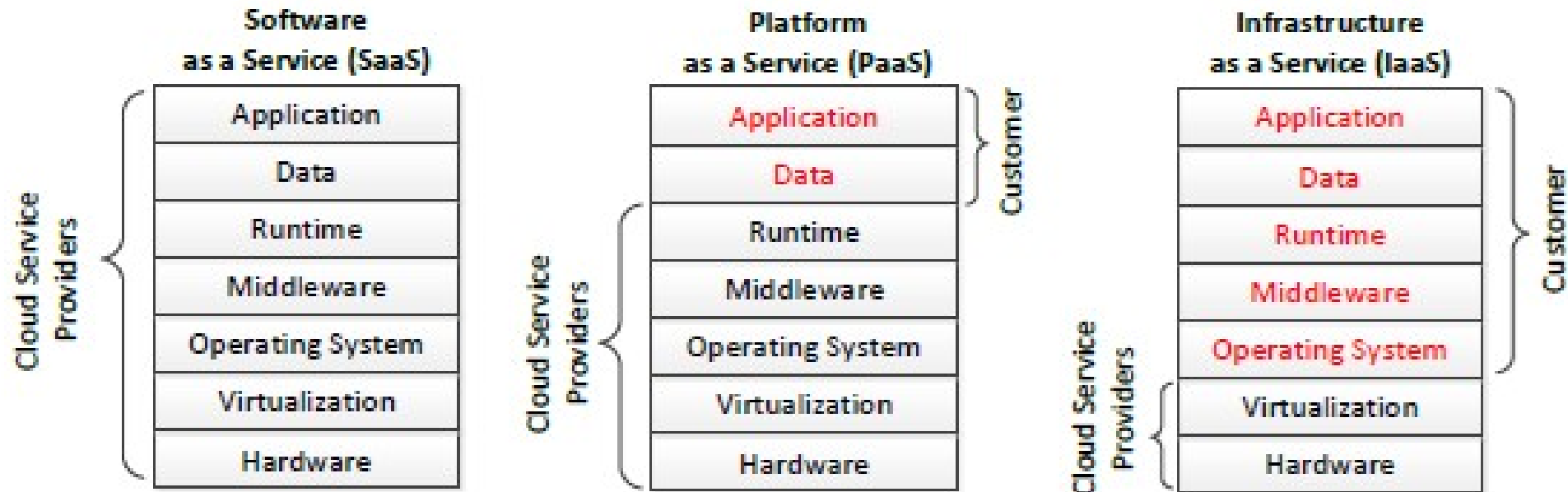


Figure 9.3: Layers of cloud computing environment owned by customer and cloud service provider on three service models: IaaS, PaaS, and SaaS (public cloud).

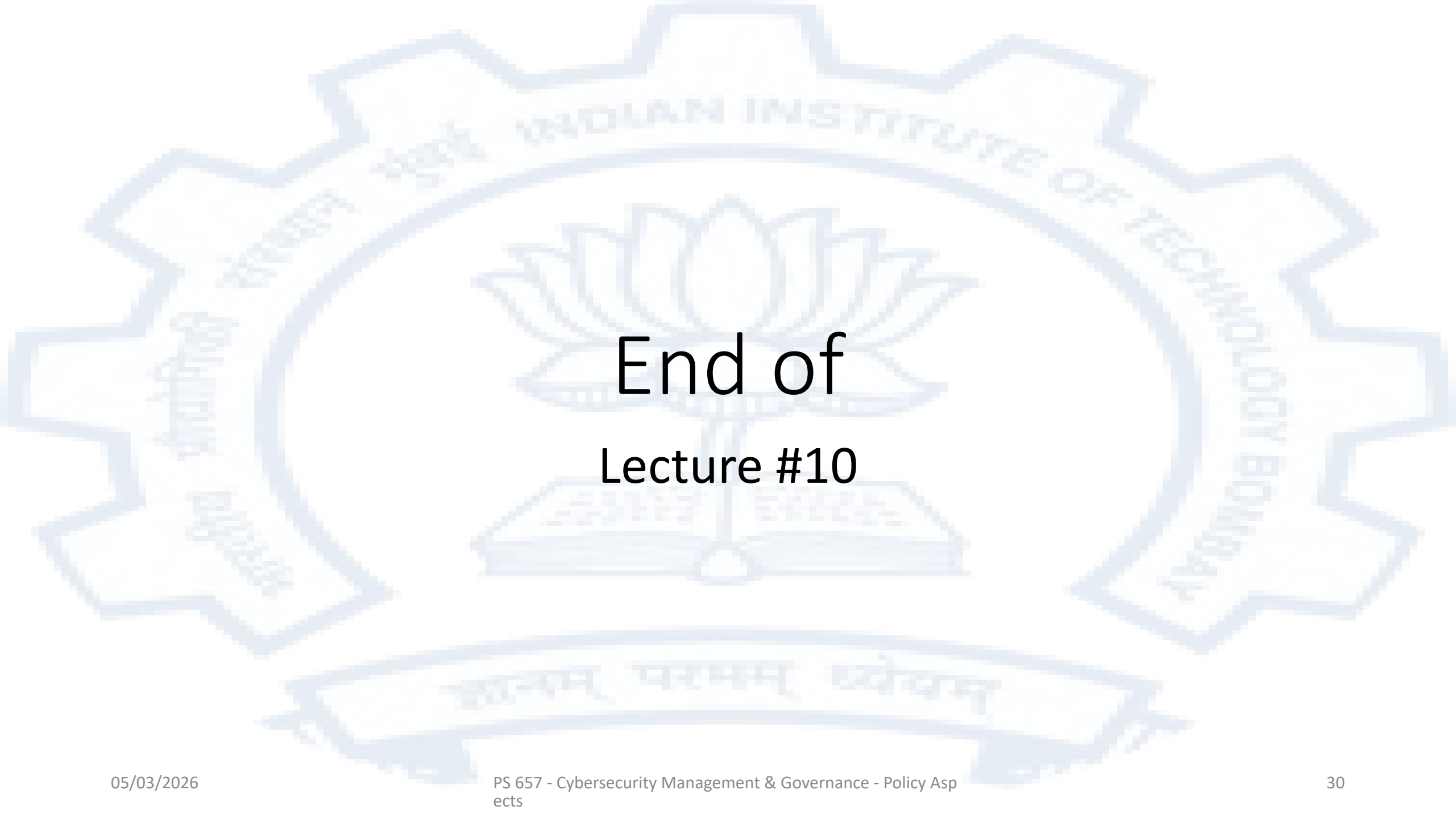
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Cloud Forensics Aspects and Challenges

- Logical data acquisition is the norm. Physical data acquisition is almost totally impractical/impossible
- The cloud is the authoritative data source and not the client device
- Logging is pervasive and therefore potentially more information is known rather than having to be deduced
- Computation is distributed with code and data being downloaded and run on a client device and then leaving almost no traces on the client device
- These aspects make many of the classical forensic techniques irrelevant and new techniques are required

To summarize

- Digital forensics identifies and reconstructs the relevant sequence of events that have led to some observed state of an IT system
- The demonstrable/provable authenticity and integrity of the data sources and the scientific grounding of the methods of analysis are vital to the admissibility of conclusions in a court of law
- Today digital forensics methods may be used to anticipate or analyse events that may not lead to a legal proceeding in a court but may be part of cyber defense operations
- The various models of cloud computing pose special forensic challenges and new methods and tool chains will be needed to address these challenges.

The background of the slide features a large, light blue watermark of the Indian Institute of Technology Bombay (IIT Bombay) logo. The logo is a gear-like shape with a lotus flower in the center. The text "INDIAN INSTITUTE OF TECHNOLOGY BOMBAY" is written along the top inner edge of the gear, and "सत्यमेव जयते" is written along the bottom inner edge. The text "End of Lecture #10" is centered over the lotus flower.

End of Lecture #10